Week 6 Assignment

Sketching an experimental design from the published literature

This is a more polished, formally written version of the exercise from Wednesday's tutorial. Render this file to PDF and submit both the `.qmd` and the rendered `.pdf` via Canvas before Week 7's lecture (Monday). See the syllabus for the late policy.

Instructions

1. Find one primary research article in your own subfield that describes an actual experiment (not a purely observational study). A review article, meta-analysis, or purely descriptive/correlative study does not qualify — you need something with an assigned treatment.
2. Write up your analysis of its design, answering all seven questions from the framework introduced this week:
   - What is the experimental unit?
   - What are the treatments, and how many levels does each factor have?
   - How many experimental units per treatment (replication)?
   - Was assignment to treatment randomized, and how?
   - Was there blocking, matching, or other control for a nuisance variable?
   - What was the response variable, and how was it measured?
   - Did the statistical analysis match the design?
3. Include the full citation for the paper you chose.
4. Propose one concrete change you would make to the design (e.g., additional blocking, more replication, a different randomization scheme) and explain in 2–3 sentences what it would buy you.

Length and format

Aim for roughly one page of prose (not counting the citation) — this is meant to be a focused, careful read of one design, not an exhaustive review of the whole paper.